

6.4.3 Radioactivity

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) radioactive decay; spontaneous and random nature of decay	M1.3
(b) (i) α -particles, β -particles and γ -rays; nature, penetration and range of these radiations (ii) techniques and procedures used to investigate the absorption of α -particles, β -particles and γ -rays by appropriate materials	
(c) nuclear decay equations for alpha, beta-minus and beta-plus decays; balancing nuclear transformation equations	
(d) activity of a source; decay constant λ of an isotope; $A = \lambda N$	Learners will also require knowledge of 5.1.4(a)
(e) (i) half-life of an isotope; $\lambda t_{1/2} = \ln(2)$ (ii) techniques and procedures used to determine the half-life of an isotope such as protactinium	PAG7
(f) (i) the equations $A = A_0 e^{-\lambda t}$ and $N = N_0 e^{-\lambda t}$, where A is the activity and N is the number of undecayed nuclei	M3.12
(ii) simulation of radioactive decay using dice	M1.3

(g) graphical methods and spreadsheet modelling of the equation $\frac{\Delta N}{\Delta t} = -\lambda N$ for radioactive decay

HSW3 Using spreadsheets to model the radioactive decay of nuclei.

M0.5, M2.5, M3.9

(h) radioactive dating, e.g. carbon-dating.